Sequences

$\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16} \ldots\ldots\right\}$

$a_n = \frac{1}{2^n}$; $n = 1, 2, 3, 4$

$a = \frac{1}{2^1}, \frac{1}{2^2}, \frac{1}{2^3}$

$a_n = \left\{\frac{\ln n}{n}\right\}$

$= \lim_{n\to\infty} \frac{\ln n}{n} = \frac{\infty}{\infty}$

LH→ $\lim_{n\to\infty} \frac{\frac{1}{n}}{1} = \frac{1}{n}$

$\therefore \lim = 0$

$\boxed{\therefore \text{lim exits / Convergent}}$

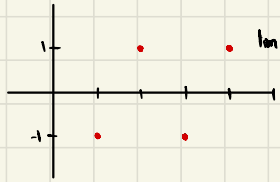
$\lim_{n\to\infty} \left\{(-1)^n\right\}$

$a_1 = (-1)^1 = -1$

$a_2 = (-1)^2 = 1$

$a_3 = (-1)^3 = -1$

$a_4 = (-1)^4 = 1$



$\lim (-1)^n = \pm 1$

2 lims Cannot Exist

$\therefore$ limit DNE

$\boxed{a_n (-1)^n \text{ is divergent}}$

Use absolute values to find limit, if limit of the absolute values is 0, the original limit is also 0, if its not 0, it is still inconclusive

$a_n = \left\{\frac{3n}{1+6n}\right\}$

$\lim_{n\to\infty} \frac{3n}{1+6n} = \lim_{n\to\infty} \frac{\frac{3n}{n}}{\frac{1}{n} + \frac{6n}{n}} = \frac{3}{0+6} = \frac{1}{2}$

$\boxed{\therefore \text{lim exists / Convergent}}$

$\lim_{n\to\infty} p^n = \begin{cases} 0 & \text{if } -1 < p < 1 \\ 1 & \text{if } p = 1 \end{cases}$ it is convergent to 0

$0 < \frac{n}{n+1} < 1$

So it is bounded

$\frac{1}{2} \quad \frac{2}{3} \quad \frac{3}{4}$

Not every monotonic sequence is Conv

But if a sequence is both bounded and monotonic it must be convergent

Series

A series is a sum of all sequences

$\sum_{n=1}^{\infty} a_n$ or $\sum a_n$

$a_n = a_1, a_2, a_3 \ldots\ldots$ Sequences

$\sum_{n=1}^{\infty} = a_1 + a_2 + a_3 \ldots\ldots\ldots$ Series

Test 1: Telescoping Series test

$\sum_{n=1}^{\infty} a_n$ $\quad S_n = \sum_{i=1}^{n} a_1 + a_2 + a_3 \ldots\ldots a_n$

$\{S_n\}$ $\quad \lim_{n\to\infty} S_n$ $\quad$ If limit exists, series $\sum_{n=1}^{\infty} a_n$ is Conv

and the limit is the overall limit

If limit does not exists, series $\sum_{n=1}^{\infty} a_n$ is div

Ex

$S_n = \sum_{i=1}^{n} \frac{1}{i(i+1)}$ $\quad \frac{1}{i(i+1)} = \frac{A}{i} + \frac{B}{(i+1)}$ $\quad A=1$, $B=-1$

$= \frac{1}{i} - \frac{1}{i+1}$

$= \sum_{i=1}^{n} \left(\frac{1}{i} - \frac{1}{i+1}\right)$

$= \left(\frac{1}{1} - \cancel{\frac{1}{2}}\right) + \left(\cancel{\frac{1}{2}} - \cancel{\frac{1}{3}}\right) + \left(\cancel{\frac{1}{3}} - \cancel{\frac{1}{4}}\right) \ldots \underbrace{\left(\cancel{\frac{1}{n-1}} - \cancel{\frac{1}{n}}\right)}_{i=n-1} + \underbrace{\left(\cancel{\frac{1}{n}} - \frac{1}{n+1}\right)}_{i=n}$

$S = 1 - \frac{1}{n+1}$

$\lim_{n\to\infty}\left(1 - \frac{1}{n+1}\right) = \lim_{n\to\infty} 1 - \lim_{n\to\infty} \frac{1}{n+1}$

$= 1 - 0$ $\quad \therefore \lim$ exists/Conv

$\frac{1}{n+1} = \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$

$\lim = 0$

$\therefore \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ Converges to 1